

EDUCATION

School of Engineering and Applied Sciences, University of Pennsylvania Master of Science in Engineering in Robotics <i>Courses: Control, Optimization and Motion Planning for Autonomous Systems; Machine Perception; Sensor Fusion and Localization</i>	Sep 2021 — May 2023 GPA: 3.87/4.0
National Institute of Technology Karnataka, Surathkal Bachelor of Technology in Electronics and Communication <i>Courses: Data Structures & Algorithms; Control Systems; Dynamical Systems; Digital System Design</i>	Aug 2016 — May 2020 GPA: 9.25/10.0

TECHNICAL EXPERIENCE

Autonomy Engineer, Luminous Robotics <i>C++, Python, ROS2 Humble, Git, Decision Making, Behavior Trees, Outdoor Autonomy, Navigation</i> - Developing the autonomy stack for a solar module installation robot, enabling automated deployment - Architected hierarchical decision-making system using Behavior Trees	Dec 2025 — Present
Robotics Engineer II, Milwaukee Tool <i>Python, C++, ROS2 Humble, Arduino, Git, Outdoor Autonomy, Planning, Control, Localization, Mapping, Manipulation, MoveIt</i> - Managed and mentored two interns, boosting team productivity and project outcomes through leadership and guidance - Leading the development of an autonomous mobile manipulator for rough terrains, focusing on navigation, localization, sensor integration, and manipulation using ROS2	Nov 2023 — Nov 2025
Research Engineer, Recupero Robotics, LLC <i>Python, Arduino, CAD, OnShape, Git, Medical Device Research, 3D printing, Microsoft Office</i> - Development and design of the existing TheraDrive robot, used for upper limb rehabilitation - Built the company website, recuperorobotics.com highlighting the product, motivation and vision	Jul 2023 — Nov 2023
Research Assistant, Rehabilitation Robotics Lab, University of Pennsylvania <i>Python, Data Analysis, CAD, OnShape, Git, Medical Device Research, 3D printing, Microsoft Office, CircuitMaker, ML</i> - Invented a novel smart sensing medical toy for collecting and classifying 4 basic infant interactions - Published "Towards an AI-driven soft toy for automatically detecting and classifying infant-toy interactions using optical force sensors" in Frontiers in Robotics and AI, Volume 11 - 2024	Nov 2021 — Nov 2023
ASIC Engineer, NVIDIA, Bangalore, IND - Performed Functional Timing Analysis and Timing Closure with PrimeTime - Debugged timing exceptions and contributed to enhancing the internal timing tool to optimize analysis by 2 times	Jul 2020 — Aug 2021

PROJECTS

SauberBOT [Video] [Report] <i>Python, C++, ROS Melodic, Git, LiDAR, Market research, Product Development, CAD, OnShape, RTK-GPS, Jetson AGX, Team size: 5</i> - Won 1st place, \$10000, for ingenuity and presentation with a market-research-backed solution - Significantly contributed to control, motion planning, and software-hardware integration of the robot	Oct 2022 — May 2023
Indoor-Outdoor Localization [Report] [Media] [Github] <i>C++, Python, ROS Melodic, Git, LiDAR, GPS, SLAM, Autonomous Vehicles, RViz, RTK-GPS, Jetson AGX, Team size: 3</i> - Successfully implemented precise indoor-outdoor localization employing advanced 3D LiDAR and HDL Graph SLAM methods - Orchestrated seamless integration of Velodyne 3D LiDAR and GPS, bolstering navigation capabilities using ROS Melodic platform	Apr — May 2023
F1 Tenth – Autonomous Racing [Media] <i>C++, Python, ROS2 Foxy, Git, LiDAR, SLAM, Autonomous Vehicles, Hardware testing, RViz, LiDAR, Jetson NX, Team size: 4</i> - Led the development and coding of high-performance control and motion planning modules - Achieved top ranking in reactive racing, follow the gap, and 3rd position in map-based racing, employing Pure Pursuit strategy	Jan — May 2023
Multi-agent Planning using Chance Constrained Model Predictive Control [Poster] [Report] [Media] [Github] <i>C++, Python, ROS2 Humble, Git, RViz, Drake, Team size: 3</i> - Implemented multi-agent motion planning and control featuring static and dynamic obstacle avoidance through non-linear MPC - Optimized trajectories by minimizing the probability of collision of uncertain robot regions, caused due to noisy localization	Nov — Dec 2022

SKILLS

Languages Python, C++, MATLAB **Platforms** Jetson, RPi **Simulation** Gazebo, RViz, Foxglove
Tools and Packages ROS, ROS2, Git, CAD, SolidWorks, OnShape, Microsoft Office, OpenCV, Drake, CircuitMaker, EDA, Docker, Linux